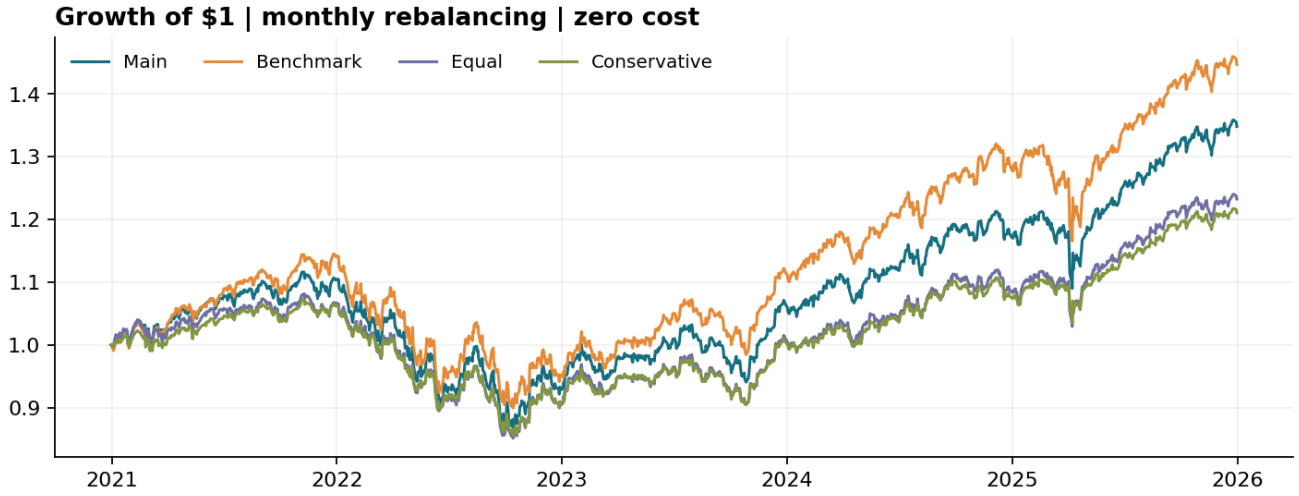
Study design and headline results

Question: how did a preset ETF portfolio, diversified across regions and bond types, perform against a US stock-bond benchmark? The weights were not chosen for the best historical result; the focus is verifiable portfolio calculations and a careful reading of the results.

Data: 2020-12-31 is the initial price anchor and returns cover 2021-01-01 to 2025-12-31, 1255 daily returns. Prices are Yahoo Finance adjusted closes, which include the vendor's dividend and split adjustments, used as a total-return proxy. Downloaded 2026-10-07 (UTC).

Portfolios

Main: VTI 30%, VB 10%, VEA 15%, VWO 5%, BND 20%, IEF 10%, TIP 5%, BNDX 5% (60% equity, 40% bonds). Benchmark: VTI 60%, BND 40%. Equal: 12.5% each. Conservative: 20%, 5%, 10%, 5%, 30%, 15%, 10%, 5% (same order). All portfolios use the same dates and monthly rebalancing.



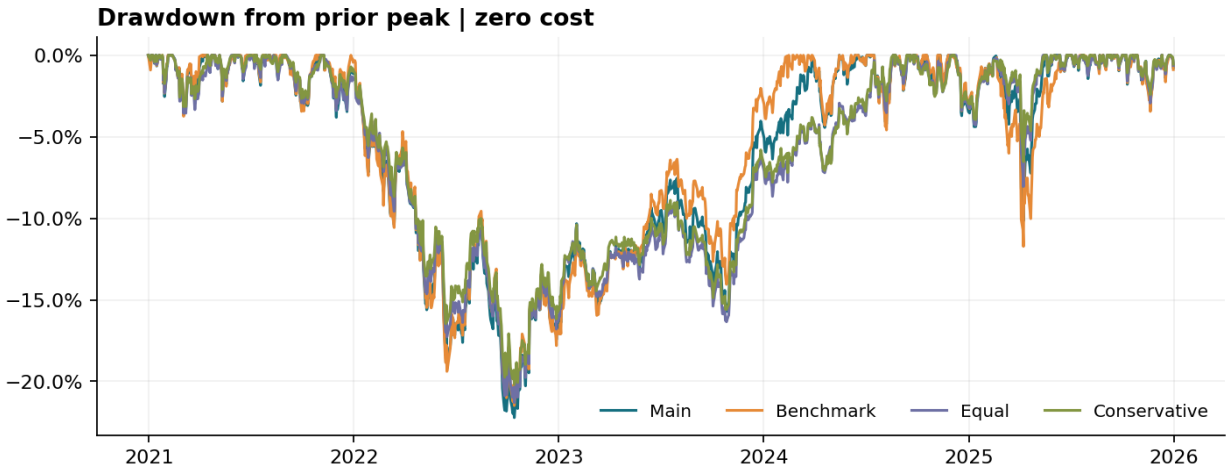
Zero-cost metrics

Portfolio	Total return	Annual return	Annual vol.	Max drawdown	Sharpe
Main	34.81%	6.18%	10.63%	-22.22%	0.62
Benchmark	44.68%	7.70%	11.18%	-21.49%	0.72
Equal	23.25%	4.29%	9.12%	-21.30%	0.51
Conservative	21.06%	3.91%	8.03%	-20.19%	0.52

Annual return compounds NAV growth over 252 trading days; volatility is the sample standard deviation of daily returns times sqrt(252). Sharpe uses daily excess returns with a zero risk-free rate (a simplifying assumption).

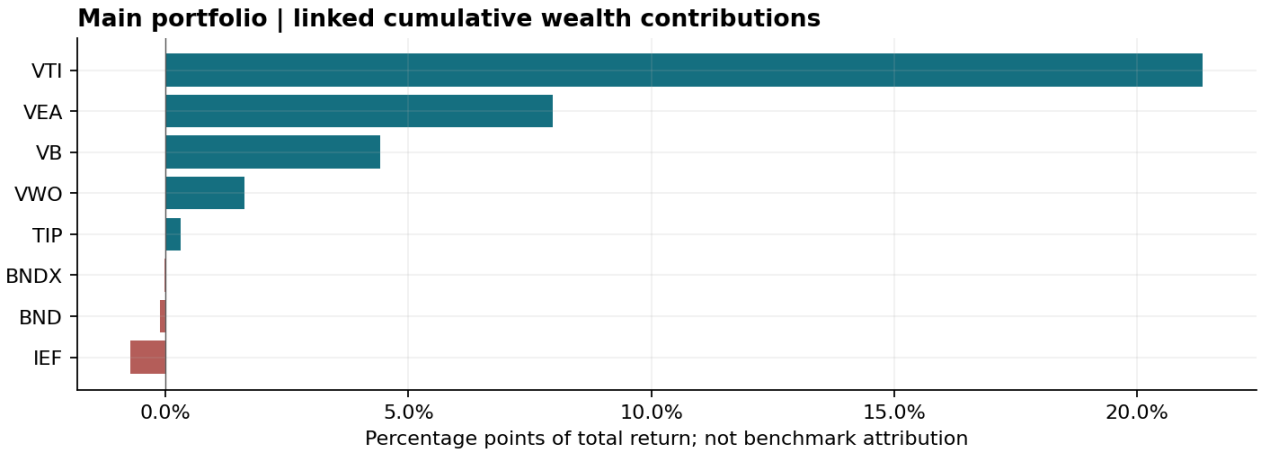
The main portfolio returned 34.81% cumulatively against 44.68% for the benchmark, a gap of -9.87 pp; the relative wealth return is -6.82%.

Risk, allocation and return contributions



Annualized volatility is 10.63% for the main portfolio and 11.18% for the benchmark; maximum drawdowns are -22.22% and -21.49%. Whether risk was lower depends on looking at both measures.

The main portfolio's worst calendar year was 2022, at -15.86% against -16.70% for the benchmark. A multi-asset allocation can still lose in the same year.



VTI made the largest cumulative wealth contribution, 21.35 pp. Contributions are linked through daily NAV and sum to the total return; they are not attribution relative to the benchmark.

In the wealth-contribution reconciliation of the return gap, VTI has the largest negative difference between its main-portfolio and benchmark contributions, -23.33 pp. This is an arithmetic reconciliation of two portfolios, not Brinson attribution or proof of cause.

Relative risk and what drives it

Tracking error is 2.27% and the information ratio -0.65. Tracking error is the annualized standard deviation of daily return differences against the benchmark; the information ratio is the annualized mean daily active return divided by tracking error.

The main portfolio holds less US equity than the benchmark and adds international equity, a small-cap tilt and other bond types, so the gap reflects these exposures. This report compares allocations and absolute return contributions only; Brinson-Fachler attribution, which splits the gap into allocation and selection effects, is in reports/v2. The equal-weight and conservative portfolios also have different stock-bond splits, which affects the risk comparison.

Linked wealth contribution is the sum of previous-day NAV times each day's asset contribution, applied to the zero-cost model only; it adds up to the cumulative total return. Daily contributions use the previous day's weights, and multi-day contribution percentages are never added directly.

Method, costs and checks

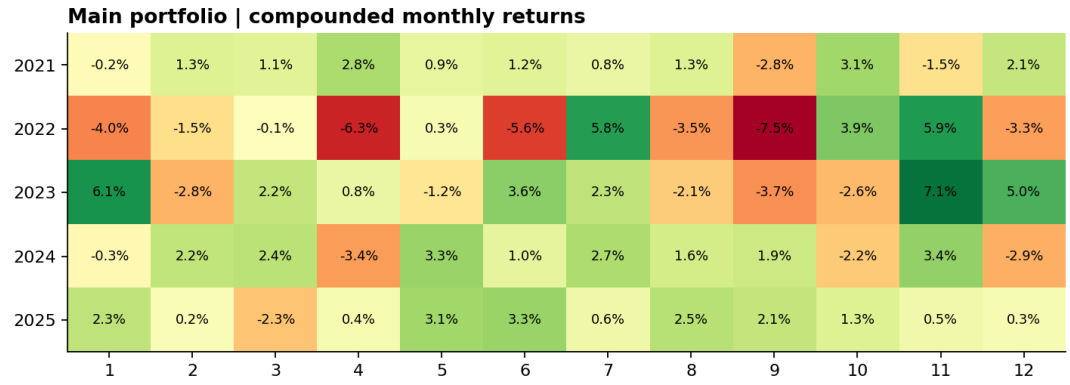
Portfolio model: r(i,t) = P(i,t)/P(i,t-1) - 1 and r(p,t) = sum of w(i,t-1) r(i,t). Between rebalances, weights drift as w(i,t) = w(i,t-1)(1 + r(i,t))/(1 + r(p,t)). Target weights are set at the month-end close and apply from the next session's return; new weights are never used on the same day.

Costs: the rate is applied to buys plus sells as a share of pre-trade NAV, the sum of |target weight - drifted weight|, not divided by two; the initial purchase counts as 100%. Costs are deducted before the next day's return: net return = (1 - cost)(1 + gross return) - 1. All holdings shrink in proportion; there is no separate cash or share ledger. There is no rebalance and no liquidation cost on the final day.

Cost sensitivity: main portfolio

Cost (bps)	Total return	Annual return	Max drawdown
0	34.81%	6.18%	-22.22%
5	34.68%	6.16%	-22.23%
10	34.54%	6.14%	-22.24%

With 10 bps costs the main portfolio returns 34.54% cumulatively, 0.27 pp less than at zero cost. Costs are charged on buys plus sells, including the initial purchase.



Checks and limitations

Checks are recorded in verification.json: a hand-computed small portfolio, an independent units-based model, SQLite LAG price returns, and consistency of NAV, returns and contributions. Automated tests cover monthly rebalancing, perturbed future prices, costs and zero denominators. The absolute tolerance is 1e-10.

Limitations: (1) adjusted prices can be revised by the vendor, and dividends and splits were not rebuilt independently; (2) taxes, slippage and trading limits are not modeled, and costs are assumptions; (3) ETF exposures overlap, and a USD listing can still carry foreign currency risk; (4) weights were fixed before the first data pull, but the study is retrospective and does not test out-of-sample performance.

Source and reproduction: Yahoo Finance historical prices; the adjustment definition is at in.help.yahoo.com/kb/adjusted-close-sln28256.html. Data are used for personal research and no redistribution license is claimed. Snapshot checksums, configuration and dependency versions are kept with the project; the README explains how to rerun it.